

SHETH L.U.J AND SIR M.V COLLEGE
SUBJECT : SCALA FOR DATA SCIENCE

MODULE 1
PRACTICAL NO - 6, 7, 8, 9, 10

Aim :

6. Slice a Breeze matrix to extract a sub-matrix and calculate its row and column sums.
7. Write a program to perform element-wise addition, subtraction, multiplication, and division of two Breeze matrices.
8. Read a CSV file and calculate basic statistics for each numeric column. Use the scala-csv library or similar tools.
9. Handle missing values in a dataset. Replace missing values with the column mean.
10. Filter rows in a dataset where a specific column value exceeds a threshold.

Practical 6

CODE :

```
import breeze.linalg._

object Practical6 {

  def main(args: Array[String]): Unit = {

    val matrix = DenseMatrix(
      (10, 20, 30, 40),
      (50, 60, 70, 80),
      (90, 100, 110, 120),
      (130, 140, 150, 160)
    )

    println("Original Matrix:")
    println(matrix)

    val subMatrix = matrix(1 to 2, 1 to 2)

    println("\nSub-Matrix:")
    println(subMatrix)

    val rowSums = sum(subMatrix(*, ::))
    println("\nRow Sums:")
    println(rowSums)

    val columnSums = sum(subMatrix(:, *))
    println("\nColumn Sums:")
    println(columnSums)

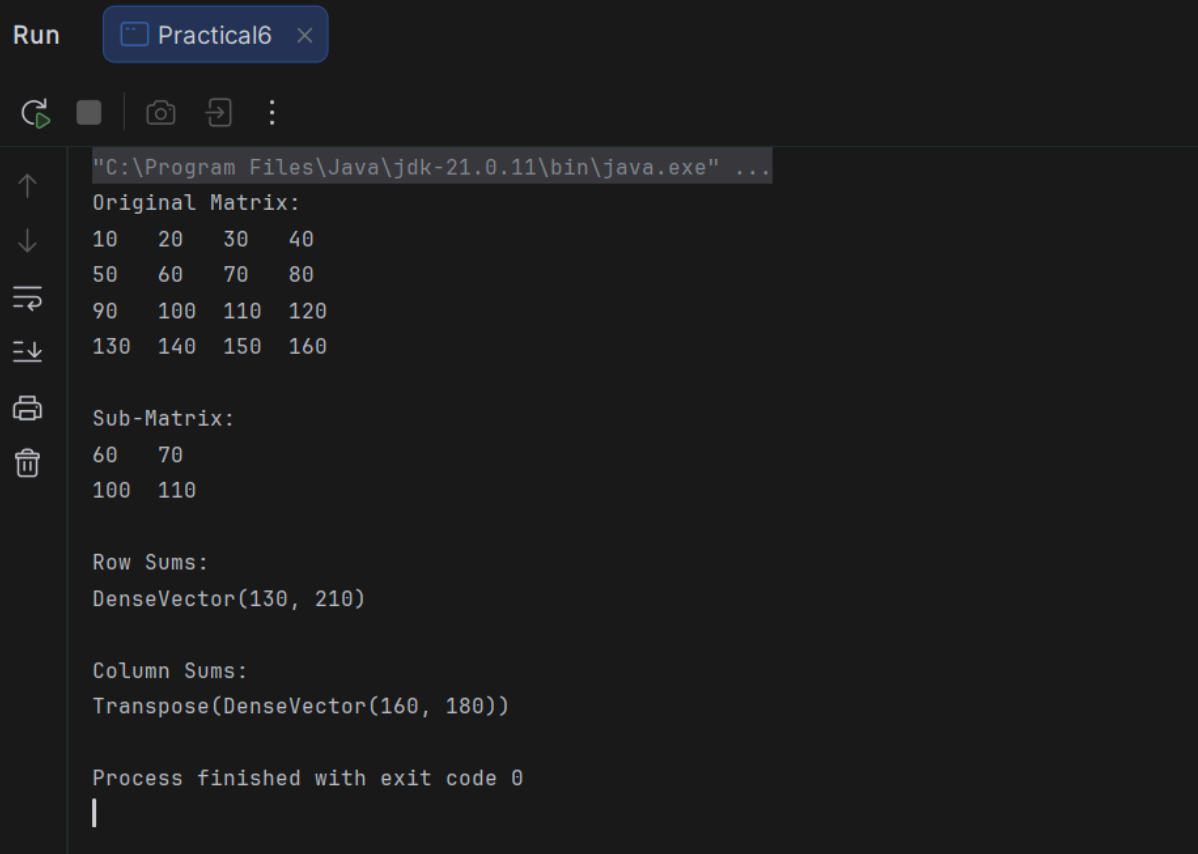
  }

}
```

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OUTPUT :



```
Run Practical6 x
C:\Program Files\Java\jdk-21.0.11\bin\java.exe ...
Original Matrix:
10  20  30  40
50  60  70  80
90  100 110 120
130 140 150 160

Sub-Matrix:
60  70
100 110

Row Sums:
DenseVector(130, 210)

Column Sums:
Transpose(DenseVector(160, 180))

Process finished with exit code 0
|
```

Practical 7

CODE :

```
import breeze.linalg._

object Practical7{

  def main(args: Array[String]): Unit = {

    val matrix1 = DenseMatrix(
      (10.0, 20.0, 30.0),
      (40.0, 50.0, 60.0),
      (70.0, 80.0, 90.0)
    )

    val matrix2 = DenseMatrix(
      (1.0, 2.0, 3.0),
      (4.0, 5.0, 6.0),
      (7.0, 8.0, 9.0)
    )

    println("Matrix 1:")
    println(matrix1)
```

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```
println("\nMatrix 2:")
println(matrix2)

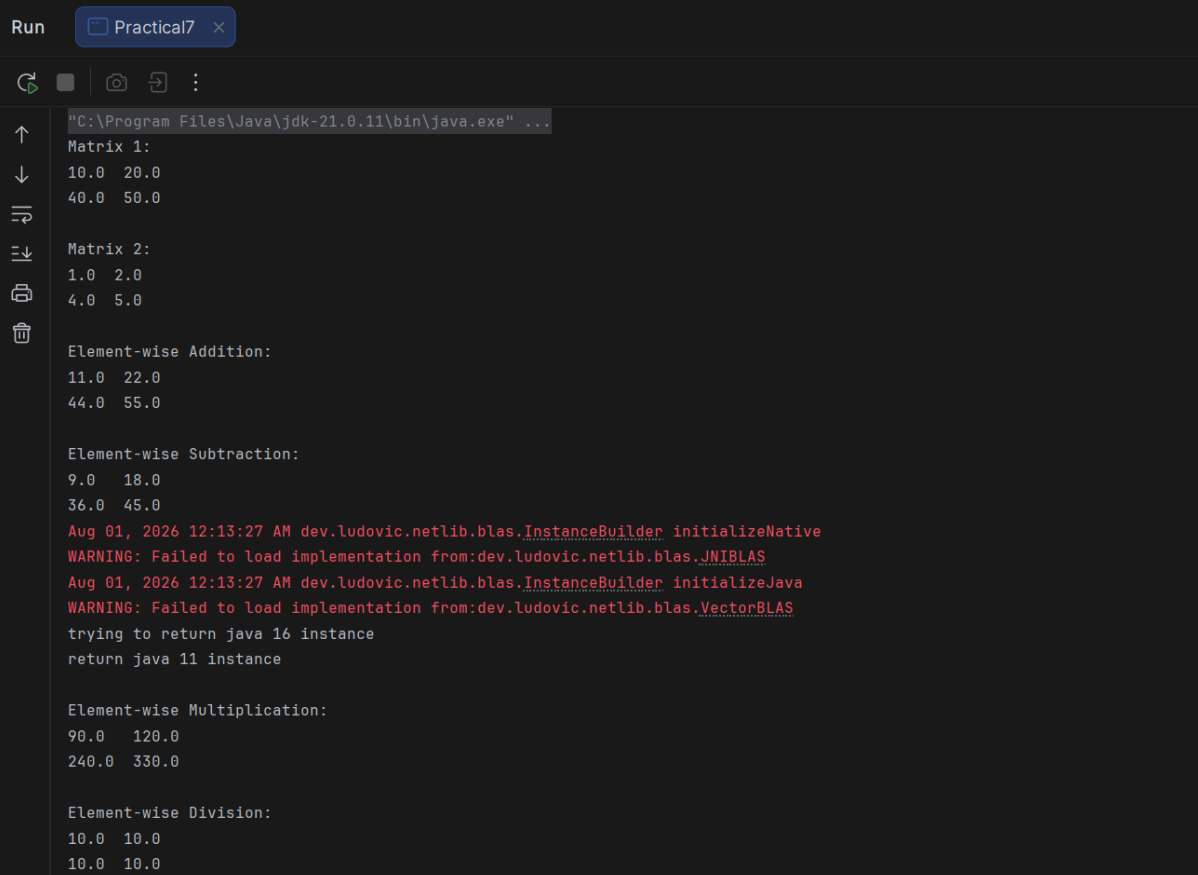
// Element-wise Addition
val addition = matrix1 + matrix2
println("\nElement-wise Addition:")
println(addition)

val subtraction = matrix1 - matrix2
println("\nElement-wise Subtraction:")
println(subtraction)

val multiplication = matrix1 * matrix2
println("\nElement-wise Multiplication:")
println(multiplication)

val division = matrix1 / matrix2
println("\nElement-wise Division:")
println(division)
}
}
```

OUTPUT :



```
Run Practical7 x
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" ...
Matrix 1:
10.0  20.0
40.0  50.0

Matrix 2:
1.0  2.0
4.0  5.0

Element-wise Addition:
11.0  22.0
44.0  55.0

Element-wise Subtraction:
9.0   18.0
36.0  45.0

Aug 01, 2026 12:13:27 AM dev.ludovic.netlib.blas.InstanceBuilder initializeNative
WARNING: Failed to load implementation from:dev.ludovic.netlib.blas.JNIBLAS
Aug 01, 2026 12:13:27 AM dev.ludovic.netlib.blas.InstanceBuilder initializeJava
WARNING: Failed to load implementation from:dev.ludovic.netlib.blas.VectorBLAS
trying to return java 16 instance
return java 11 instance

Element-wise Multiplication:
90.0   120.0
240.0  330.0

Element-wise Division:
10.0   10.0
10.0   10.0
```

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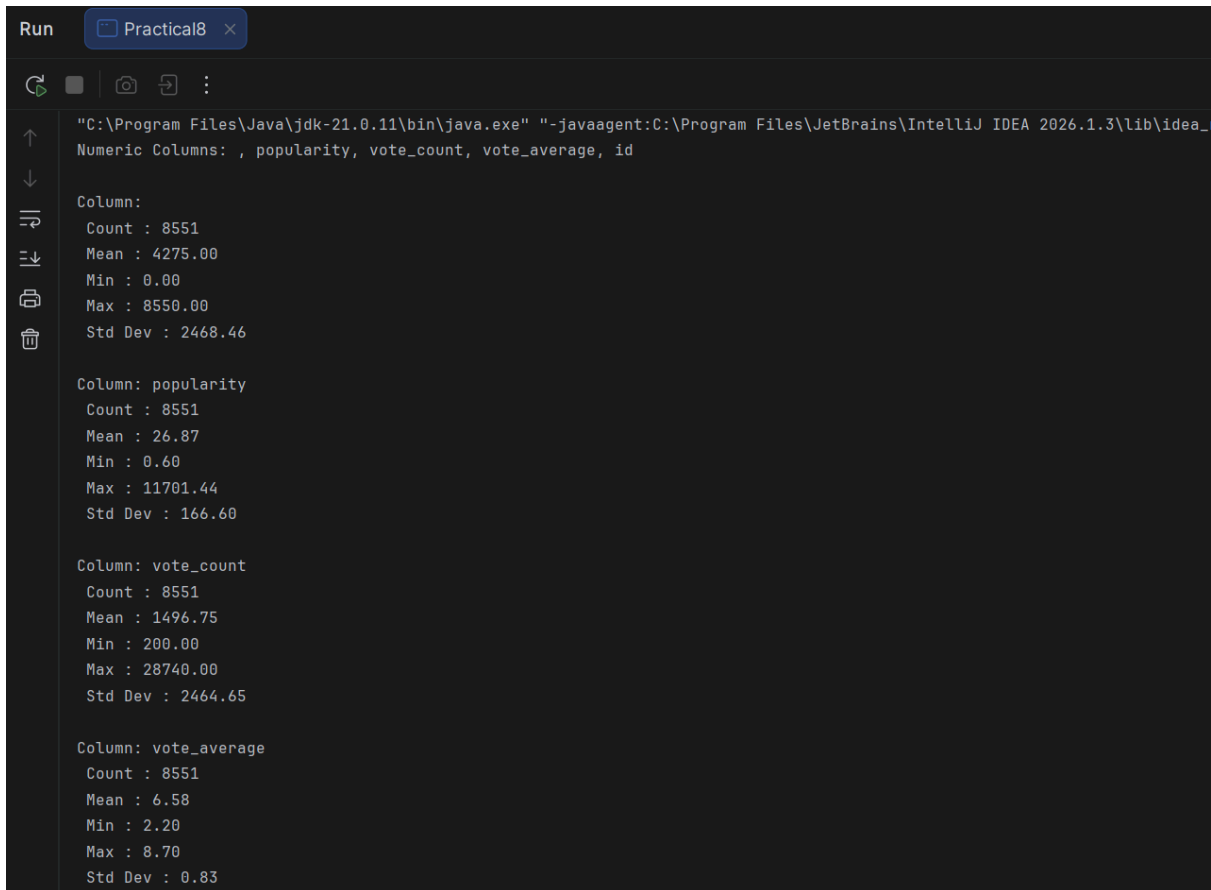
Practical 8

CODE :

```
import com.github.tototoshi.csv._
import java.io.File
import scala.util.Try
import scala.math._
object Practical8 {
  def main(args: Array[String]): Unit = {
    val reader = CSVReader.open(new File("movies.csv"))
    val allRows = reader.allWithHeaders()
    reader.close()
    val numericCols = allRows.head.keys.filter { col =>
      Try(allRows.head(col).toDouble).isSuccess
    }.toList
    println(f"Numeric Columns: ${numericCols.mkString(", ")}\n")
    for (col <- numericCols) {
      val values = allRows.flatMap(row =>
        Try(row(col).toDouble).toOption)
      val count = values.size
      val mean = values.sum / count
      val minVal = values.min
      val maxVal = values.max
      val stdDev = sqrt(values.map(x => pow(x - mean, 2)).sum / count)
      println(s"Column: $col")
      println(f" Count : $count")
      println(f" Mean : $mean%.2f")
      println(f" Min : $minVal%.2f")
      println(f" Max : $maxVal%.2f")
      println(f" Std Dev : $stdDev%.2f\n")
    }
  }
}
```

OUTPUT :

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The screenshot shows the Run console of IntelliJ IDEA with a tab labeled 'Practical8'. The console output displays the following information:

```
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_...
Numeric Columns: , popularity, vote_count, vote_average, id

Column:
Count : 8551
Mean : 4275.00
Min : 0.00
Max : 8550.00
Std Dev : 2468.46

Column: popularity
Count : 8551
Mean : 26.87
Min : 0.60
Max : 11701.44
Std Dev : 166.60

Column: vote_count
Count : 8551
Mean : 1496.75
Min : 200.00
Max : 28740.00
Std Dev : 2464.65

Column: vote_average
Count : 8551
Mean : 6.58
Min : 2.20
Max : 8.70
Std Dev : 0.83
```

Practical 9

CODE :

```
import com.github.tototoshi.csv._
import java.io.File
import scala.util.Try
object omwala_practical9 {
  def main(args: Array[String]): Unit = {
    val inputFile = new File("Player_Attributes.csv")
    val reader = CSVReader.open(inputFile)
    val allRows = reader.allWithHeaders()
    reader.close()
    val numericColumns = Seq("overall_rating", "potential")

    val stats: Map[String, (Double, Int)] = numericColumns.map { col =>
      val values = allRows.map(row => row.getOrElse(col, "").trim)
      val validNumbers = values.flatMap(v => Try(v.toDouble).toOption)
      val missingCount = values.count(v => Try(v.toDouble).isFailure)
      val mean = if (validNumbers.nonEmpty) validNumbers.sum / validNumbers.size
      else 0.0
      (col, (mean, missingCount))
    }.toMap
```

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```
println("\n--- Missing Data Report ---")
stats.foreach { case (col, (mean, missingCount)) =>
  println(f"Column: $col, Missing values: $missingCount, Replaced with mean:
$mean%.2f")
}

val cleanedRows = allRows.map { row =>
  numericColumns.foldLeft(row) { (accRow, col) =>
    val value = accRow.getOrElse(col, "").trim
    val replaced = Try(value.toDouble).toOption match {
      case Some(_) => value
      case None => f"${stats(col)._1}%.2f"
    }
    accRow.updated(col, replaced)
  }
}

val outputFile = new File("titanic_cleaned.csv")
val writer = CSVWriter.open(outputFile)
val headers = cleanedRows.head.keys.toSeq
writer.writeRow(headers)
cleanedRows.foreach(row => writer.writeRow(headers.map(row)))
writer.close()
println("\n Missing values replaced and saved to omwala_cleaned.csv")
}
```

OUTPUT :



```
Run omwala_practical9 x
C:\Program Files\Java\jdk-21.0.11\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\i
--- Missing Data Report ---
Column: overall_rating, Missing values: 836, Replaced with mean: 68.60
Column: potential, Missing values: 836, Replaced with mean: 73.46

Missing values replaced and saved to omwala_cleaned.csv

Process finished with exit code 0
```

Practical 10

CODE :

```
import com.github.tototoshi.csv._
import java.io.File
object omwala_practical10 {
  def main(args: Array[String]): Unit = {
    val reader = CSVReader.open(new File("Player_Attributes.csv"))
```

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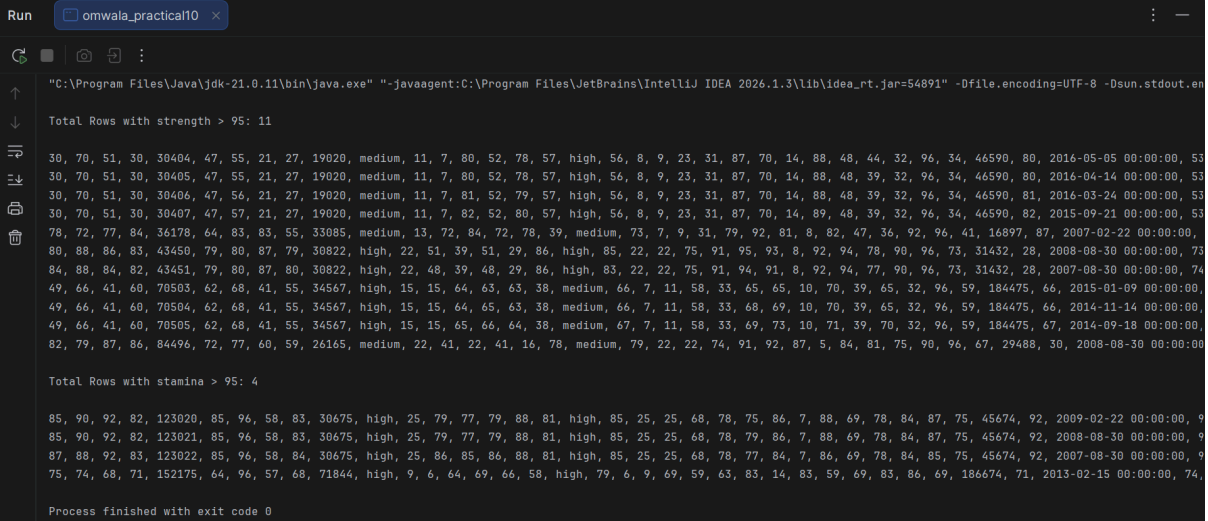
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```
val data = reader.allWithHeaders()
reader.close()
val threshold = 95
```

```
val filteredRows1 = data.filter { row =>
  row.get("strength").exists(value => value.toIntOption.exists(_ >
    threshold))
}
println(s"\nTotal Rows with strength > $threshold: ${filteredRows1.length}\n")
filteredRows1.foreach { row =>
  println(row.values.mkString(", "))
}
```

```
val filteredRows2 = data.filter { row =>
  row.get("stamina").exists(value => value.toIntOption.exists(_ >
    threshold))
}
println(s"\nTotal Rows with stamina > $threshold: ${filteredRows2.length}\n")
filteredRows2.foreach { row =>
  println(row.values.mkString(", "))
}
}
```

OUTPUT :



```
Run omwala_practical10 x
"C:\Program Files\Java\jdk-21.0.11\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.3\lib\idea_rt.jar=54891" -Dfile.encoding=UTF-8 -Dsun.stdout.en

Total Rows with strength > 95: 11

30, 70, 51, 30, 30404, 47, 55, 21, 27, 19020, medium, 11, 7, 80, 52, 78, 57, high, 56, 8, 9, 23, 31, 87, 70, 14, 88, 48, 44, 32, 96, 34, 46590, 80, 2016-05-05 00:00:00, 53
30, 70, 51, 30, 30405, 47, 55, 21, 27, 19020, medium, 11, 7, 80, 52, 78, 57, high, 56, 8, 9, 23, 31, 87, 70, 14, 88, 48, 39, 32, 96, 34, 46590, 80, 2016-04-14 00:00:00, 53
30, 70, 51, 30, 30406, 47, 56, 21, 27, 19020, medium, 11, 7, 81, 52, 79, 57, high, 56, 8, 9, 23, 31, 87, 70, 14, 88, 48, 39, 32, 96, 34, 46590, 81, 2016-03-24 00:00:00, 53
30, 70, 51, 30, 30407, 47, 57, 21, 27, 19020, medium, 11, 7, 82, 52, 80, 57, high, 56, 8, 9, 23, 31, 87, 70, 14, 89, 48, 39, 32, 96, 34, 46590, 82, 2015-09-21 00:00:00, 53
78, 72, 77, 84, 36178, 64, 83, 83, 55, 33085, medium, 13, 72, 84, 72, 78, 39, medium, 73, 7, 9, 31, 79, 92, 81, 8, 82, 47, 36, 92, 96, 41, 16897, 87, 2007-02-22 00:00:00,
80, 88, 86, 83, 43450, 79, 80, 87, 79, 30822, high, 22, 51, 39, 51, 29, 86, high, 85, 22, 22, 75, 91, 95, 93, 8, 92, 94, 78, 90, 96, 73, 31432, 28, 2008-08-30 00:00:00, 73
84, 88, 84, 82, 43451, 79, 80, 87, 80, 30822, high, 22, 48, 39, 48, 29, 86, high, 83, 22, 22, 75, 91, 94, 91, 8, 92, 94, 77, 90, 96, 73, 31432, 28, 2007-08-30 00:00:00, 74
49, 66, 41, 60, 70503, 62, 68, 41, 55, 34567, high, 15, 15, 64, 65, 63, 38, medium, 66, 7, 11, 58, 33, 65, 65, 10, 70, 39, 65, 32, 96, 59, 184475, 66, 2015-01-09 00:00:00,
49, 66, 41, 60, 70504, 62, 68, 41, 55, 34567, high, 15, 15, 64, 65, 63, 38, medium, 66, 7, 11, 58, 33, 68, 69, 10, 70, 39, 65, 32, 96, 59, 184475, 66, 2014-11-14 00:00:00,
49, 66, 41, 60, 70505, 62, 68, 41, 55, 34567, high, 15, 15, 65, 66, 64, 38, medium, 67, 7, 11, 58, 33, 69, 73, 10, 71, 39, 70, 32, 96, 59, 184475, 67, 2014-09-18 00:00:00,
82, 79, 87, 86, 84496, 72, 77, 60, 59, 26165, medium, 22, 41, 22, 41, 16, 78, medium, 79, 22, 22, 74, 91, 92, 87, 5, 84, 81, 75, 90, 96, 67, 29488, 30, 2008-08-30 00:00:00, 74,

Total Rows with stamina > 95: 4

85, 90, 92, 82, 123020, 85, 96, 58, 83, 30675, high, 25, 79, 77, 79, 88, 81, high, 85, 25, 25, 68, 78, 75, 86, 7, 88, 69, 78, 84, 87, 75, 45674, 92, 2009-02-22 00:00:00, 9
85, 90, 92, 82, 123021, 85, 96, 58, 83, 30675, high, 25, 79, 77, 79, 88, 81, high, 85, 25, 25, 68, 78, 79, 86, 7, 88, 69, 78, 84, 87, 75, 45674, 92, 2008-08-30 00:00:00, 9
87, 88, 92, 83, 123022, 85, 96, 58, 84, 30675, high, 25, 86, 85, 86, 88, 81, high, 85, 25, 25, 68, 78, 77, 84, 7, 86, 69, 78, 84, 85, 75, 45674, 92, 2007-08-30 00:00:00, 9
75, 74, 68, 71, 152175, 64, 96, 57, 68, 71844, high, 9, 6, 64, 69, 66, 58, high, 79, 6, 9, 69, 59, 63, 83, 14, 83, 59, 69, 83, 86, 69, 186674, 71, 2013-02-15 00:00:00, 74,

Process finished with exit code 0
```

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